

1. Scientific Literature



Papers &
Databases



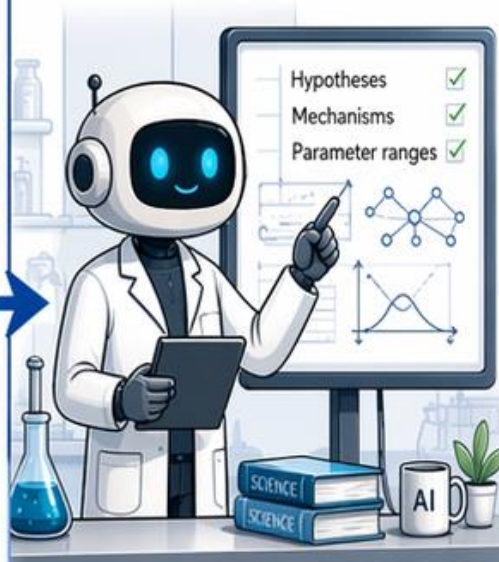
Search &
Query



Knowledge
Extraction

unstructured text $\rightarrow$ knowledge

2. Agent



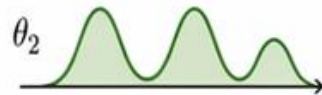
searches literature



constructs informative priors

3. Informative Priors

Prior distributions over
latent parameters
 $p(\text{latent parameters})$



...



4. BOED Planner

choose next design
(next experiment)



controllable design variables

$$\mathbf{x} \in \mathcal{X}$$

e.g., $x_1, x_2, \dots, x_m$

arbitrary design spaces

next
experiment

5. Scientific Simulator / Experiment (arbitrary)

Inputs

Controllable
design variables

$$\mathbf{x} \in \mathcal{X}$$



Latent / Unknown
(parameters)

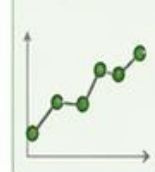
$$\theta \in \Theta$$



arbitrary simulators
(black-box or known models)

Output

Observations /
Outputs
 $\mathbf{y} \in \mathcal{Y}$



6. Posterior Update (Belief Update)

update beliefs over latent parameters

Prior
 $p_t(\theta)$

$\mathbf{y}_t$

Posterior
 $p_{t+1}(\theta)$



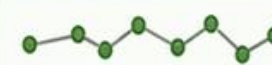
Bayesian update with new observations

Agent in the loop

continuously refines priors,
hypotheses, and parameter
ranges from new evidence
and literature.

Observations / Outputs

$$\mathbf{y}_t$$



Key Concepts

- ✓ arbitrary simulators
- ✓ arbitrary priors
- ✓ arbitrary design spaces
- controllable (known to set)
- latent (unknown to learn)
- output / observations



Literature & Agent
(knowledge acquisition)



Priors & Inference
(beliefs over latents)



BOED & Planning
(decision making)



Simulator / Experiment
(arbitrary systems)



Observations & Update
(learning & feedback)